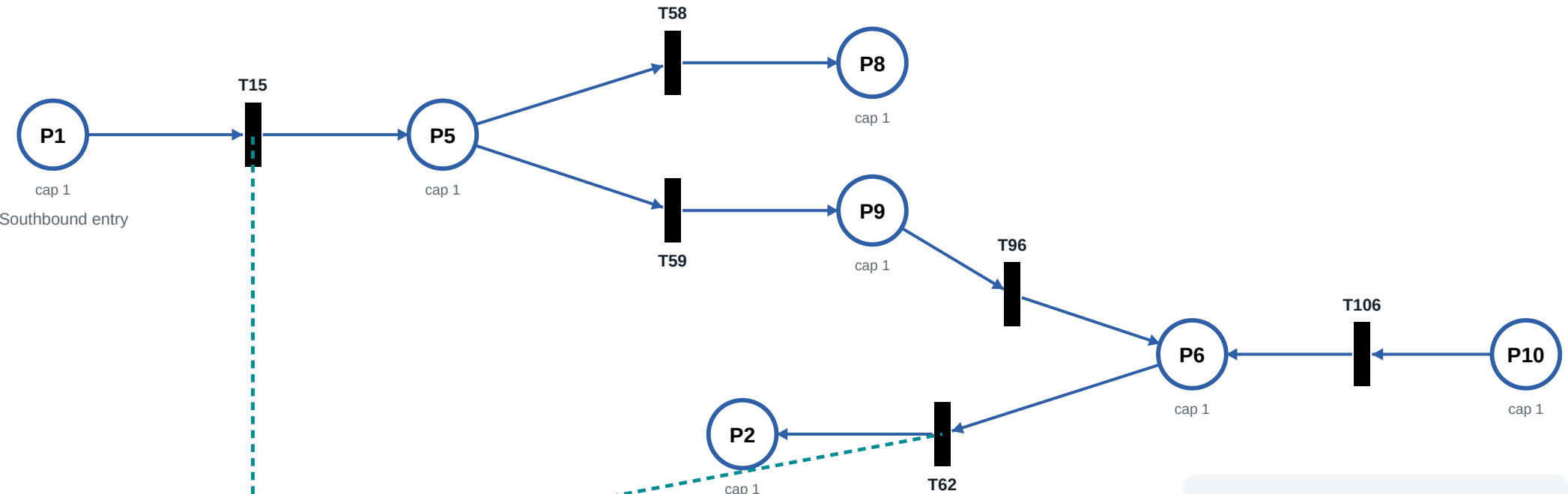


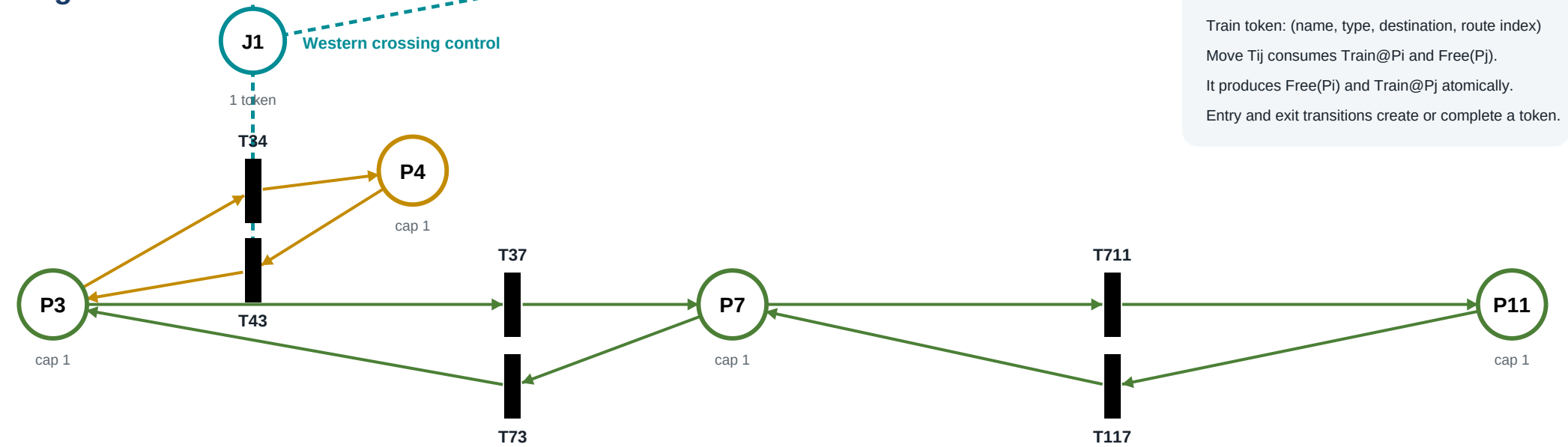
Railway interlocking Petri-net

Coloured train tokens; every track-section place has capacity one

Passenger subnet



Freight subnet



Token and firing rule

Train token: (name, type, destination, route index)

Move Tij consumes Train@Pi and Free(Pj).

It produces Free(Pi) and Train@Pj atomically.

Entry and exit transitions create or complete a token.

Transition guards and invariants

Rules used to turn the Petri-net design into an atomic Java movement step

Type	Entry	Destination	Section route	Enabled when
Passenger south	1	8	1 - 5 - 8	Next section is free; western crossing lock is available for T15.
Passenger south	1	9	1 - 5 - 9	Next section is free; opposite use of the 5-9-6 junction is not selected.
Passenger north	9	2	9 - 6 - 2	Next section is free; T96 is serialised against T59.
Passenger north	10	2	10 - 6 - 2	Next section is free.
Freight south	3	4	3 - 4	Section 4 is free and neither T15 nor T62 is selected.
Freight south	3	11	3 - 7 - 11	Next section is free.
Freight north	4	3	4 - 3	Section 3 is free and neither T15 nor T62 is selected.
Freight north	11	3	11 - 7 - 3	Next section is free.

Safety and progress properties

One-safety	For every section s, occupancy(s) is either 0 or 1.
Type separation	Passenger tokens never enter freight places, and freight tokens never enter passenger places.
Atomic movement	All candidates are checked against the same snapshot before any approved origin is cleared.
Collision exclusion	One target has at most one approved incoming move; edge swaps and crossing conflicts do not fire together.
Passenger priority	An enabled passenger crossing transition suppresses T34 and T43 for that movement round.
Fair contention	Waiting rounds rank otherwise equal candidates, with request order used only as a final tie-break.
Initial marking	Free(P1)..Free(P11) = 1, J1 = 1, and all Train@Pi places are empty.